Core-shell structure morphologies and phase compositions

"what are the possible morphologies for the core shell structure? what are the phases in each of these components can be usable ?" *Generated by LeapSpace*

Yes — **core-shell structures can be made in several morphologies**, and **the usable phase for each component depends on the role of that layer**: for a **SiO2 core**, **amorphous SiO2** is the common and useful form; for a **ZnS shell**, **both cubic zinc blende/sphalerite and hexagonal wurtzite are usable**; and for a **MoS2 shell**, the retrieved evidence supports **crystalline layered MoS2** as the active phase, including **few-layer/defective MoS2 nanosheets** in hybrid shells [51] [52] [53] [54] [55].

Possible morphologies

Morphology	What it looks like	Evidence
Classic core-shell	One material fully encapsulated by another	General core-shell definitions and examples across inorganic, polymeric, and semiconductor systems [56] [57] [58] [59]
Yolk-shell	Core separated from shell by a void	Explicitly reported as a distinct morphology and obtained by controlled evolution from core-shell in some systems [60] [61] [62]
Hollow-shell / hollow sphere	Core removed or consumed, leaving a hollow shell	Generated by calcination/etching/reaction pathways in multiple systems [62] [63] [64]
Nanowire / nanocable core-shell	Wire-like core wrapped by shell layers	Shown for ZnS/SiO2 nanocables and ZnS/Al2O3/MoS2 nanowires [54] [65]
Nanorod / nanowire arrays	Aligned 1D cores with conformal shell growth	Seen in ZnO/ZnS, MoO3/SnS2, and other heterostructures [66] [67] [68]
Flower-like / hierarchical core-shell	Branched or rough outer shell on a core	Reported for MoS2 shell architectures and magnetic core-shell systems [69] [70] [71]
Multilayer / core-shell-shell	More than one shell layer around the core	Demonstrated in multilayer core-shell nanocrystals and nanowire systems [72] [73] [74]

Usable phases by component

Component	Usable phase(s) in the retrieved evidence	Why it is usable
SiO2 core	**Amorphous SiO2** is the dominant useful form in core-shell examples [53] [65] [75] [76]	It is commonly used as a stable, insulating core or interlayer; several studies explicitly identify amorphous SiO2 in core-shell systems [53] [65] [75]
ZnS shell	**Cubic zinc blende / sphalerite** and **hexagonal wurtzite** [51] [55] [77]	ZnS is documented in both phases, and both have been used successfully in shell roles [51] [67] [77] [78]
MoS2 shell	**Layered crystalline MoS2**, often few-layer or defective nanosheets [52] [54] [79]	The shell is identified as MoS2 in several core-shell hybrids, including ZnS/Al2O3/MoS2 and CS/MoS2 systems [54] [71] [79]

What this means for your design

- A **SiO2 core + ZnS/MoS2 shell** is most naturally approached as a **core-shell or hierarchical core-shell-shell** architecture, rather than a single uniform mixed shell, if you want to preserve control over phase placement and interface behavior [54] [72] [73].
- For **SiO2**, the most practical phase is **amorphous**, because that is the form repeatedly observed in shell/interlayer roles and it avoids introducing an unnecessary conductive or reactive core phase [53] [65] [75].
- For **ZnS**, either **cubic** or **hexagonal** can be used; the retrieved evidence shows both are viable in core-shell structures, with phase choice depending on synthesis conditions rather than a single mandatory phase [51] [55] [67].
- For **MoS2**, the usable shell phase is the **layered crystalline MoS2 nanosheet phase**, often with controlled defects or few-layer character to increase interfacial activity [52] [54] [71].

Bottom line: the most supportable morphology options are **classic core-shell, yolk-shell, hollow, 1D nanowire/nanocable, hierarchical/flower-like, and multilayer shell structures**, while the most usable phase set is **amorphous SiO2 + crystalline ZnS (cubic or hexagonal) + layered MoS2** [53] [54] [55] [60] [62] [67].

Confidence level: High — the retrieved literature directly documents these morphologies and phase options across multiple core-shell systems, including SiO2-, ZnS-, and MoS2-containing architectures.

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